HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed backend services in **Python** and **Java Spring Boot** to ingest, process, and store data from IoT devices and sensors for a connected home platform.
- Designed and implemented **REST APIs** and interfaces for seamless data exchange between physical product prototypes, cloud services, and user-facing applications.
- Built interactive **dashboards** using **Power BI** to visualise product and sensor data, transforming raw information into actionable insights for innovation teams.
- Explored emerging software and **IoT technologies**, including MQTT and WebSockets, to identify new opportunities in life safety and environmental sensing.
- Experimented with **AI development tools** to accelerate software development, coding, and rapid prototyping of new digital capabilities, reducing iteration cycles by approximately **30%**.
- Containerised applications using **Docker** and deployed them to cloud environments, managing configurations and ensuring predictable rollouts for proof-of-concept systems.
- Collaborated with cross-functional teams, including electronics and embedded engineers, to integrate physical prototypes with digital platforms and applications, ensuring cohesive system functionality.

SKILLS

Languages – Python, Java, SQL, JavaScript, TypeScript

Frameworks & Libraries – Spring Boot, React, NestJS, Flask, FastAPI, pandas, NumPy, scikit-learn

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Databases – PostgreSQL, MySQL, Oracle, NoSQL

Tools & Technologies – Docker, Kubernetes, Jenkins, Git, Jira, Splunk, Dynatrace, Power BI, Tableau, R/RStudio, MQTT, WebSockets, AI Development Tools

Methodologies – Agile, Scrum, CI/CD, Data Analytics, Machine Learning, IoT

PROJECTS

Connected Home Sensor Data Platform

- Engineered a scalable backend using **Python** and **FastAPI** to collect and process real-time data streams from a fleet of 500+ connected home sensors via **MQTT**.
- Developed **RESTful APIs** for data access and integrated with a PostgreSQL database, enabling efficient querying and analysis of sensor readings.
- Built a **Power BI dashboard** to visualise environmental data (temperature, humidity, air quality), providing key insights for product development teams and reducing data exploration time by **40%**.

AI-Assisted Prototyping Tool

- Developed a proof-of-concept application using **Python** and **Flask** to demonstrate AI's potential in accelerating software development, leveraging AI coding assistants.
- Integrated a language model API to assist with code generation and debugging tasks, improving the speed of initial prototype development by an estimated **25%**.
- Designed a simple UI using basic HTML/JavaScript to interact with the AI tools, enabling rapid testing of generated code snippets and design ideas.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)